

4.4

Method of undetermined coefficients

Equations

(1) Non-homogenous ODE has a RHS called the forcing function.

$$ay'' + by' + cy = f(t)$$

(2) A nice function is below. This is required for using undetermined coeffs.

$$\begin{aligned} f(t) &= Ct^m e^{\alpha t} \\ \text{or} \\ f(t) &= Ct^m e^{\alpha t} (\sin \beta t + \cos \beta t) \end{aligned}$$

(3) The particular solution.

- The solution obtained from non-homogenous analysis
- It is also the guess that we plug into the ODE
- If LHS has sin or cos then y_p must have both in the guess. see (6)

$$\begin{aligned} y_p &= t^s (A_m t^m + \dots + A_1 t + A + 0) e^{\alpha t} \\ y_p &= t^s (A_m t^m + \dots + A_1 t + A) e^{\alpha t} \cos \beta t + t^s (B_m t^m + \dots + B_1 t + B) e^{\alpha t} \sin \beta t \end{aligned}$$

(4) The general solution:

- It is the sum of the solution obtained from the homogenous equation, plus that from our non-homogenous solution, aka the particular.
- This does represent every y which satisfies our ODE.
- y_h is 0 when we plug into our ode, so thats why we solve y_p and y_h separately, then combine.

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$$y = \underbrace{y_h}_{\text{homogenous}} + \underbrace{y_p}_{\text{particular}}$$

Methods:

We need to get our guess for the [particular-solution](#) (3)

1. **Solve the homogenous** if it wants general solution. Else just get the roots r_1 and r_2 .
2. Check how many times you see the roots in $f(t)$. The variable s is equal to that number.
3. The variable m is equal to highest degree in $f(t)$
4. The variables r and/or α are also given by $f(t)$

- If you have a sum of forcing functions, then use superposition
- The if you have even one trig function, then you *must* assume that $y(p)$ is both sin and cos. see (6)

Examples

(5)

$$\begin{aligned} y'' + 3y' + 2y &= 10e^{3t} \\ y_p &= \frac{1}{2}e^{3t} \end{aligned}$$

(6) - Try by guessing only sin or only cos

$$\begin{aligned} y'' + 3y' + 2y &= \sin(t) \\ y_p &= A \cos t + B \sin t \\ y'_p &= -A \sin t + B \cos t \\ y''_p &= -A \cos t - B \sin t \end{aligned}$$

After plugging into ode group the sin and cos:

$$\begin{aligned} \sin t &= (-A = 3B + 2A) \cos t + (-3B - 3A + 2B) \sin t \\ \sin t &= (A + 3B) \cos t + (B - 3A) \sin t \\ \implies B - 3a &= 0 \end{aligned}$$

And from there you solve to get

$$A = .1, B = -.3$$

(7)

$$\begin{aligned} y'' + 4y &= 5t^2 e^t \\ 5t^2 e^t &= e^t \underbrace{(2A_1 + 2A_2 + A_3 + 3A_3)}_{=0} + t e^t \underbrace{(4A_1 + A_1 + 4A_2)}_{=0} + t^2 e^t \underbrace{(A_1 + 4A_1)}_{=5} \\ A_1 = 1A_2 &= -\frac{4}{5}, A_3 = -\frac{2}{25} \\ y_p &= \left(t^2 - \frac{4}{5}t - \frac{2}{25} \right) e^t \end{aligned}$$

(8) First one where y_h and y_h collide

$$y'' + y' = 5$$

Note that $r_1 = 0, r_2 = 1$ and that $m = 0$ because $5t^0 = 5$

$$y_p = At = 5t$$

(9)

$$y'' - 6y' + 9y = e^{3t}$$

$$r_1 = r_2 = 3, r = 3 \implies s = 2$$

$$\begin{aligned} y_p &= t^2 e^{3t} \\ y'_p &= 2te^{3t} + 3t^2 e^{3t} \\ y_p &= [2e^{3t} + 6te^{3t}] + [6te^{3t} + 9t^2 e^{3t}] \end{aligned}$$

and then the solution is :

$$A = \frac{1}{2}, y_p = \frac{1}{2}t^2 e^{3t}$$